Chapter 3

Enhancing Customer Service with Conversational AI Chatbots

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The use of AI chatbots represents a major shift away from conventional customer service to a more effective approach. Previously, customer support relied solely on interactions between customers and staff. Now, using AI chatbots, companies can automate tasks, reduce costs, and provide round-the-clock support. In this chapter, we look at conversational AI (CAI) chatbots, which are the next iteration of traditional chatbots.

These advanced CAI chatbots can handle a wide range of customer service tasks. In some cases, they've already shown that they can outperform staff when it comes to consistency and availability. This chapter looks at the transformational role of CAI chatbots for customer service and the impact they can have on your business and customer satisfaction.

Finding Out about Conversational AI Chatbots

Customer service chatbots are AI-powered programs that handle conversations with customers across communication channels. They can answer questions, provide information, and help with tasks like taking orders or assisting with account issues.

Understanding the process

Traditional rules-based chatbots provided limited responses and no ability to learn from customer interactions. CAI chatbots use natural language processing (NLP) and machine learning (ML) to meet customer needs.

It's useful to understand the capabilities that enable CAI chatbots to deliver automated services. They can

- Understand the context of the response to respond appropriately during a customer interaction
- Process personalized data like order history to provide customized responses
- Escalate issues that involve human emotions or very nuanced language to human agents
- Integrate across backend systems (like servers or databases) to efficiently serve customers

Here's how a chatbot technically "responds" to a customer question:

1. The customer submits a query to the chatbot. It could be text, voice, or another input method.
2. The query is captured and preprocessed by the CAI system. The CAI system analyzes the request to identify the user's intent, which involves processing NLP techniques to understand the meaning.
3. The chatbot matches the customer intent to predefined categories in the chatbot's knowledge base. This information helps the chatbot understand the context of the request.
4. That information is used to generate a response that addresses the customer's query. The response can be text, images, audio, or a combination of these formats, depending on what the company chooses.
5. The generated response is delivered back to the customer.

Defining the differences between conversational AI chatbots and ChatGPT

To understand chatbots further, we need to understand the differences between CAI chatbots and ChatGPT. Unfortunately, some people incorrectly label chatbots as ChatGPT. When it comes to customer service, this can be confusing and misleading. CAI chatbots and ChatGPT both use AI, but they differ in their capabilities and use cases — especially when it comes to handling conversations.

Developers designed CAI chatbots for use cases like customer service or sales support. You can integrate them into things like apps or messaging platforms (for example, Facebook Messenger) to answer questions or help with transactions. They focus on mimicking human-like conversation using AI. They combine NLP and ML to create a more natural interaction.

Although you can combine CAI chatbots with GenAI capabilities from tools such as ChatGPT and Microsoft Copilot, the two technologies are not the same thing. CAI chatbots are developed to predictably respond to questions about a limited number of topics. GenAI chatbots are multipurpose tools designed to generate original content that's similar to content it was trained on. See Chapter 4 of this book to find out how to use Microsoft Copilot Studio to create a CAI that uses capabilities from Microsoft Copilot.



REMEMBER CAI chatbots are customizable for the specific needs of a business or application. You can integrate them with backend systems and databases to provide responses based on real-time data. For example, CAI chatbots can retrieve corporate data like purchase history to answer questions about shipping dates.

Visualizing the differences

Visualizing the differences between CAI chatbots and ChatGPT can be helpful in understanding them. Table 3-1 details the clear distinctions.

TABLE 3-1 Conversational AI Chatbots versus ChatGPT

Feature	*Conversational AI Chatbots*	*ChatGPT (Advanced AI)*
Operation	AI-driven, typically using a mix of rules and ML techniques	Utilizes deep learning (GPT architecture)
Conversation handling	Capable of handling a variety of interactions with some level of contextual understanding	Capable of handling a broader range of topics, more contextually relevant
Flexibility	More flexible than traditional chatbots; can adapt to different conversation styles	Highly adaptable, generates responses based on volumes of training data
Learning capability	Some ability to learn from interactions, but is limited	Doesn't learn in real-time; relies on pretraining data
Context understanding	Better at handling off-script conversations than traditional chatbots but has limitations	Better at understanding context but not adept at highly complex dialogues
Human-like interaction	More adept at mimicking human conversation than traditional chatbots	Highly capable of mimicking human-like conversation styles



REMEMBER CAI chatbots blend rule-based approaches with AI elements. This provides improved quality over traditional chatbots but is not at the level of sophistication of ChatGPT.

Benefitting from Conversational AI Chatbots for Customer Service

CAI chatbots provide immediate responses. This quick response leads to higher customer satisfaction and a better overall service experience. You can get several other benefits by deploying CAI chatbots:

- **The ability to provide 24/7 support:** CAI chatbots are operational all day, every day. This constant availability ensures that customer support is accessible at any time, serving customers in different time zones and schedules.
- **Cost efficiency:** By handling routine inquiries and tasks, CAI chatbots can make large customer service teams unnecessary. This leads to savings in operational costs over time.
- **Scalability:** CAI chatbots can manage a high volume of queries at one time without compromising the quality of the service. This scalability is helpful for businesses experiencing growth or seasonal spikes in customer inquiries.
- **Task automation:** By automating repetitive tasks, CAI chatbots free up human agents to focus on higher-priority customer issues. This division of labor increases overall efficiency in customer service operations.
- **Improved data collection:** CAI chatbots are effective in gathering customer data during interactions. You can use this information to improve future marketing strategies and service offerings to make them more in tune with customer needs.
- **Reduced wait times:** CAI chatbots significantly cut down the time customers spend waiting for responses. This efficiency is key to maintaining customer satisfaction and reducing the frustration associated with long wait times.
- **Help for internal staff:** Although chatbots assist customers, they're also valuable to internal staff. Having chatbots deal with internal duties provides the following advantages:
  - **Reduction in burnout:** By taking over the handling of routine inquiries, chatbots can help prevent agent burnout. Agents can stay focused on delivering better service instead of jumping back and forth from issue to issue.
  - **Improved efficiency:** With chatbots handling the usual questions, agents can use their time to handle tasks that require judgment. This leads to a

more efficient workflow and helps to improve their performance.

- **Reduced ticket volume:** By resolving basic issues using automation, the use of chatbots reduces the number of tickets that agents need to handle.

CONVERSATIONAL AI CHATBOTS VERSUS TRADITIONAL AUTOMATED RESPONSE SYSTEMS

Customer service has taken a great leap forward with the introduction of CAI chatbots. Previously, traditional automated response systems couldn't deliver a satisfying customer experience. CAI chatbots offer a more seamless experience because of their ability to understand natural language dialogue instead of looking for specific words.

Traditional response systems were rigid and frustrating to use, especially if the customers asked questions that were outside programmed responses. Here are three key differences that demonstrate the value of CAI chatbots over traditional systems:

- **Learning and adaptability:** CAI chatbots utilize ML and NLP to learn from interactions, adapt to new queries, and improve over time. They understand and respond to a range of natural language inputs. Traditional response systems typically worked based on preprogrammed rules and set responses. They couldn't learn from or adapt based on past interactions and were limited to expected questions.
- **Conversational capabilities:** CAI chatbots are designed to simulate human conversations and understand context. Traditional automated response systems were limited to basic, menu-driven interactions (like phone tree systems). They couldn't handle open-ended conversations.
- **Personalization:** CAI chatbots can provide personalized responses based on customer data and past interactions. When using traditional response systems, personalization was limited, and these systems typically couldn't store or process specific information beyond each session.

The advances that CAI chatbots provide are a great improvement over the old way. The difference in customer experience is recognized and welcomed by customers.

Constructing Conversational AI Chatbots

When it comes to implementing chatbots for customer service, several key points should shape your development. Here are several considerations to use to guide your chatbot development:

- **Finding the right use cases for your chatbot:** It is crucial to identify the specific areas in which chatbots can add value to customer service. It could be handling frequently asked questions (FAQs), providing product recommendations, or providing order tracking. Make sure the use case aligns with customer needs.
- **Defining the chatbot's purpose:** Before building a chatbot, you need to define its purpose. This involves determining what tasks it will do, the questions it can answer, and the tone it should use.

  REMEMBER Setting a chatbot tone that matches your brand is important. Creating a tone that feels natural to the brand will make your customers feel comfortable using it.
- **Focusing on customer-centric design:** Always keep the customer at the forefront when you're designing your chatbot. Make sure you know their pain points and preferences.
- **Making data-driven decisions:** Leverage data and analytics to improve your chatbot's performance. Monitor customer interactions and use feedback to identify areas for improvement.

Measuring the Return on Investment of Conversational AI Chatbots

One question frequently asked by management regarding chatbots is about the return on investment (ROI) from implementing them. Understanding how to measure the ROI of CAI chatbots is key to justifying their use. In the next sections, we look at some important measures you can take and let you know the results you should look for.



REMEMBER Before calculating ROI, it's important to set a baseline for comparison. This helps evaluate the direct impact of chatbots on business operations and your customers' experiences. You should also continuously monitor relevant metrics to assess the ongoing effects on your business.

Improving customer service efficiency

To do: Compare the efficiency and costs between human agents and CAI chatbots in handling customer inquiries.

Expected outcome: Look for a reduction in average handling time and an increase in queries resolved without human intervention. However, for the full picture, consider the complexity of queries that chatbots can handle and their evolving capabilities.

Reducing operational costs

To do: Analyze the reduction in staffing, training, and overhead expenses. Include costs related to integrating the chatbot into existing systems and the required updating of its knowledge base.

Expected outcome: There may be a reduction in the need for a large customer service team.



REMEMBER Before making staff adjustments, remember to factor in initial setup costs and ongoing maintenance for your CAI chatbot.

Boosting sales conversion rates

To do: Track the sales conversions initiated by chatbot interactions. Consider the design, interaction quality, and integration of the chatbot into the sales process.

Expected outcome: CAI chatbots can typically upsell or cross-sell, which directly contributes to the bottom line. Be aware that the effectiveness can vary by industry and application.

Enhancing lead generation

To do: Assess the chatbot's effectiveness in generating and qualifying leads. Look at the quality of leads and the efficiency of the qualification

process.

Expected outcome: Successful lead generation by chatbots is a value-add to sales teams.



REMEMBER The integration of chatbots into the overall sales strategy is the key to this success. You need to have your sales team on board, working in conjunction with the technology.

Tracking engagement metrics

To do: Examine interaction patterns, including interaction frequency, duration, and repeat usage. Measure engagement subjectively and consistently.

Expected outcome: You should expect increased engagement and customer satisfaction levels, but methods of measurement should be carefully considered.



TIP Use customer feedback on chatbot interactions to get more insights about customer sentiment. You can capture this information without having to quiz customers directly, which is a great advantage.

Increasing resolution rate

To do: Calculate the percentage of queries completely resolved by the chatbot. Consider the chatbot's capability to escalate complex issues to human agents.

Expected outcome: You should get an increase in resolution rates and customer satisfaction levels. Just remember to consider the complexity of resolved questions.

Ensuring accuracy

To do: Monitor the accuracy and misunderstanding rates of chatbot responses — factor in the chatbot's learning and adaptation capabilities over time.

TRAVELING ON AMTRAK WITH CONVERSATIONAL AI

Betting on AI to solve business problems turned out to be a great investment for Amtrak, the largest train passenger provider in the United States. According to a 2023 LinkedIn article written by Ethel Emmons (`www.linkedin.com/pulse/how-ai-chatbot-transforming-customer-service-amtrak-ethel-emmons`), Amtrak made impressive gains using a CAI chatbot for customer service.

- **Problem:** Amtrak experienced a 47.4 percent decrease in passengers in 2020 due to the pandemic (with a slight recovery in 2022), but the numbers were still well below peak ridership.
- **Solution:** Amtrak invested in an AI-backed chatbot, "Next IT," evolving from its initial "Julie" system created in 2012. The AI chatbot offered personalized, 24/7 customer support, improving customer satisfaction and engagement. This new AI chat platform handles more than five million queries annually, ranging from travel bookings to general inquiries, significantly improving customer service efficiency.
- **Results:** The AI system reported a 50 percent annual increase in usage, contributing to a 30 percent increase in revenue compared to other booking methods. Amtrak saved approximately $1 million annually in customer service expenses, with an 800 percent ROI.

This demonstrates how even traditional companies can realize significant gains using AI.

Expected outcome: High accuracy rates will mean better ROI. The CAI chatbot's ability to learn and improve from interactions is a factor in long-term ROI.

Integrating Conversational AI Chatbots into Existing Systems

The key to integrating AI chatbots into existing company systems is balancing automation with human oversight. To successfully integrate chatbots, here are a few steps to take:

- **Assess conversational workflows.** Take a close look at your current conversational processes between your customers and agents. Identify areas where chatbots can automate certain steps to streamline the workflow.
- **Ensure data integration.** Determine how the chatbot will integrate with your existing data sources. Plan which data the chatbot will need to access to serve customers, such as account information or order history.
- **Develop NLP components.** Create intents, entities, and dialogs to enhance the chatbot's natural language capabilities. Use existing support content, such as FAQs and documentation, to speed up the development process.

  REMEMBER Developers teach chatbots to understand language by building three main parts:
  - **Intents:** What the customer is trying to do (for example, book a flight).
  - **Entities:** The details about what the customer wants. These are variables that the user needs to fill in to help the chatbot understand and get details from the user's message.
  - **Dialogs:** Possible conversations around each intent. These are different ways that customers may state their requests.

  This allows the chatbot to understand the requests, collect the details, and respond to customer questions.
- **Evaluate channel options.** Consider which channels are most appropriate for your chatbot, such as website, text message, or social media direct messages (DMs). Integrate the primary channels that your customers commonly use first.
- **Plan for scalability.** Expect that conversation volume will increase over time. Make sure that your chatbot is designed to handle increased usage, especially on your popular channels.
- **Prioritize maintainability.** Regular content and feature updates are necessary, so your chatbots must be easy to update and improve.
- **Address security concerns.** Consider security, compliance, and data privacy requirements. Understand the policies that need to be followed because chatbots have access to customer data.



TIP Consider enabling human handoff workflows for conversations the AI can't handle well. Combining AI and human support can ensure

round-the-clock coverage.

Personalizing Customer Interactions

CAI chatbots are designed to deliver personalized experiences — light years away from the generic, script-based responses that were previously available. These chatbots improve customer support by providing highly personalized assistance.

For example, if you had a specific issue in the past, the CAI chatbot can recall those details, making the resolution process more efficient. Each interaction with the chatbot then becomes a building block that creates a more personalized customer experience.



TIP CAI chatbots can enhance the customer journey by "remembering" preferences and past interactions. This allows for quicker transactions and support and improves the customer experience. (For more about the customer journey, check out Chapter 1 in this book.)

Improving the shopping experience

When it comes to online shopping, CAI chatbots have transformed the experience by making shopping more personalized. They can

- Personalize shopping advice based on browsing history
- Offer product recommendations customized to individual customer preferences
- Assist with the checkout processes, making it more likely that customers won't abandon their shopping carts before making a purchase

Ensuring ethical personalization

The ability of chatbots to remember past interactions and preferences raises privacy concerns. You must design and operate them in compliance with data protection regulations and with respect for customer privacy.



TIP For more information about using AI responsibly and ethically, turn to Book 1, Chapter 6.

Using Chatbots with Human and AI Collaboration

Choosing between live agents and chatbots for customer service is a major decision. It's tempting to choose a pure human agent system or a chatbot-only model for all your customer-facing interactions. But it's important to understand that both options have their limitations, so you may want to consider a hybrid model.



TIP A hybrid model combines the efficiency of chatbots with the personal attention of live agents. It's effective because it adapts to the specific needs of each customer. Some people may want quick, automated responses, while others want human interaction.

Understanding the hybrid model

In a hybrid system, the opening interaction is typically handled by a chatbot. CAI chatbots answer FAQs and perform basic tasks. Chatbots are very efficient, but they're not yet capable of handling very complex questions. This is where the live agent steps in, and the user is informed that they've been sent to a human.

Over time, as live agents interact with the system, they contribute to the chatbot's "learning." This is done using ML, in which the chatbot gradually improves its responses and decision-making capabilities by analyzing new data. The more the chatbot is used, the smarter it becomes, eventually leading to higher performance output.

Collaborating with agents and chatbots

Agents can monitor chatbot conversations in real time. This can prove useful because it's not just about human interaction but also about learning and improving the chatbot. If the chatbot struggles or a customer shows signs of frustration, the agent can take over. This transition should be natural, with the chatbot programmed to understand signs of customer unhappiness and suggest escalating to a live agent.



TIP Are you wondering whether a hybrid customer service system raises questions about staff availability for a 24/7 operation? You don't have to have round-the-clock staffing by live agents. You have the option of using chatbots exclusively for non-business hours. If your business is global, you can shift to staff members who are working at different hours from different locations. Also, if a chatbot encounters a question it can't handle and no agents are available, it can let the customer know about the next available time for live support and offer to schedule a callback or email response.

Considering Best Practices

Although using advanced chatbots for customer service is relatively new, there are several best practices you can follow:

- **Simplify design.** Plan for clear interactions in your chatbot. Use simple language instead of jargon to avoid confusing your customers. The key is to provide easy communication.
- **Incorporate visuals.** Where appropriate within a conversation, integrate visual elements like images, graphics, and buttons. You can use these visuals to explain a point further or facilitate customer input. It's more appealing than heavy blocks of text alone.

  REMEMBER Make sure visual aspects meaningfully supplement the information. Everything you display should improve the clarity of the conversation.
- **Gather feedback.** Provide customers with an easy mechanism to offer feedback, such as a simple thumbs up/down or number rating. This gives you the

ability to constantly track satisfaction. It also helps you identify issues and add popular features.

- **Create a personality.** Give your chatbot a personality that reflects your brand's voice. Use humor (carefully) to make interactions more fun. Make your chatbot feel like an extension of your brand voice. Make sure to keep the personality consistent across features so the conversations don't feel robotic.
- **Stay up to date with trends.** Your chatbot content can't remain the same over time. Things like language and priorities are constantly changing. You need to update phrasing or even web slang to stay current. You want to demonstrate that you understand your customers and speak their language.
- **Identify escalation triggers.** Determine pain points or topics in customer issues early on that provide signals for automated or customer-driven escalation to live agents as needed. You should create these triggers based on such things as contextual cues, perceived lack of confidence in the chatbot responses, or direct requests for agents. The goal is to maximize how often the chatbot can resolve the customer's needs without getting human help by using triggers.



TIP *Escalation triggers* are rules that transfer a customer from the CAI chatbot to a human agent when additional help is needed. This happens if the customer asks for an agent or if the chatbot doesn't understand a question. The triggers detect when the AI chatbot reaches its limits in helping someone. All chatbots have some gaps in their abilities. The triggers bridge the human/AI capabilities.

- **Conduct A/B testing.** Continuously perform A/B testing to perfect the chatbot's responses and flow. This lets you improve the answers by testing strategies and using the most successful ones. Continuous experimentation using A/B testing of conversation components provides the feedback you need to optimize performance.

Reviewing Options for Creating Chatbots

Not every business has the in-house technical expertise or resources necessary to implement CAI chatbots. This may be because you have a lack of specialized knowledge in AI or limited access to the tools. You also have maintenance and updates to keep up with. If you're in this position, you can

- **Use a no-code chatbot platform.** If your team doesn't have coding knowledge, you may want to consider a no-code platform. These platforms typically have

drag-and-drop interfaces and prebuilt templates. They're limited in the ways you can customize them, but if you want to try this option, here are tools to consider:

  - **Manychat (`https://manychat.com`):** Manychat, shown in Figure 3-1, has customizable templates to build bots without coding skills.
  - **Chatfuel (`https://chatfuel.com`):** Chatfuel is easy to customize to create bots for messaging apps visually.
  - **Microsoft Copilot Studio (`www.microsoft.com/en-us/microsoft-copilot/microsoft-copilot-studio`):** Copilot Studio enables you to build custom AI agents (see Chapter 4) and custom Copilot plugins (see Chapter 5).

- **Outsource chatbot development.** You can hire agencies or freelancers for chatbot design. This gives you access to expert skills if you don't have technical staff. It also enables you to get a chatbot customized to your exact needs. Of course, this would typically cost more than no-code solutions.
- **Utilize a prebuilt chatbot.** You may want to use customizable chatbot templates made for specific industries or functions. This saves time and gets you started. The chatbot should meet your needs right out of the box, but it may not integrate easily with your other internal systems.
- **Adopt chatbot plugins or integrations.** You may try integrating chatbot plugins into existing website platforms or your customer relationship management (CRM) system. This helps streamline operations and integrate with workflows. You should also be aware that this makes you reliant on third-party program updates.
- **Utilize chatbots with limited functionality.** You can implement chatbots for specific tasks like answering FAQs or handling customer inquiries. This simplifies development and management and is faster to build. However, you may find it difficult to expand its scope in the future.
- **Explore managed chatbot services.** If you want to focus on things like content creation and branding, consider outsourcing the technical management of your chatbots. This would allow you to have your builds and updates managed for you. The caveat is that you're then dependent on a vendor's limitations and timeline. If you want to try this option, here are some tools that manage backend operations:
  - **LivePerson (`www.liveperson.com`):** LivePerson, is a CAI platform that combines chatbots and human agents.
  - **Chatbotly (`https://chatbotly.co`):** Chatbotly, shown in Figure 3-2, is a managed chatbot creation and hosting platform that handles the building, deploying, and updating of bots.



TIP Start with a chatbot pilot or trial. Beginning on a small scale allows you to experiment and reduce risks.

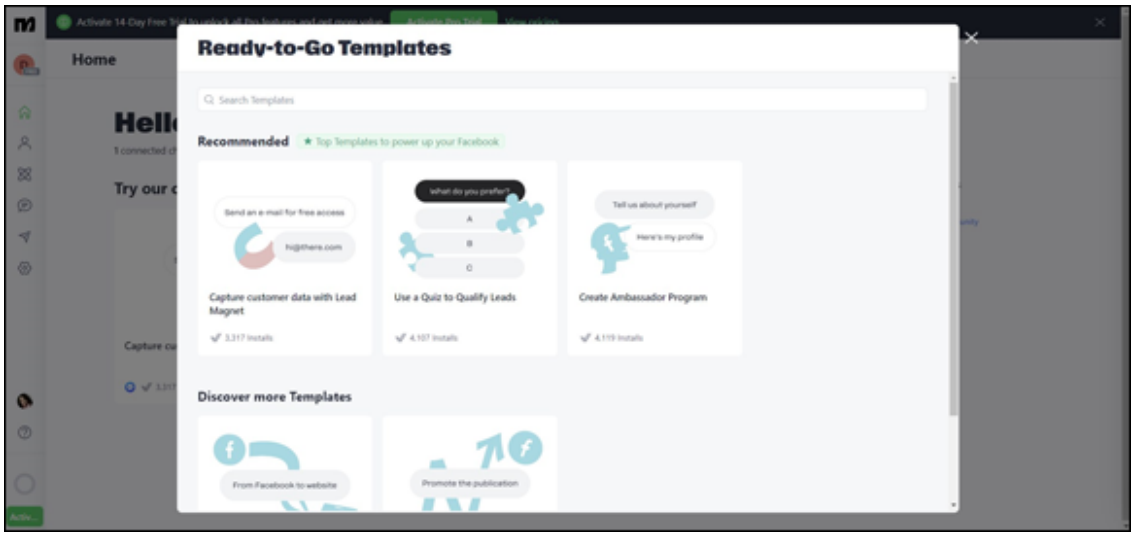


FIGURE 3-1: Manychat has many templates to choose from.

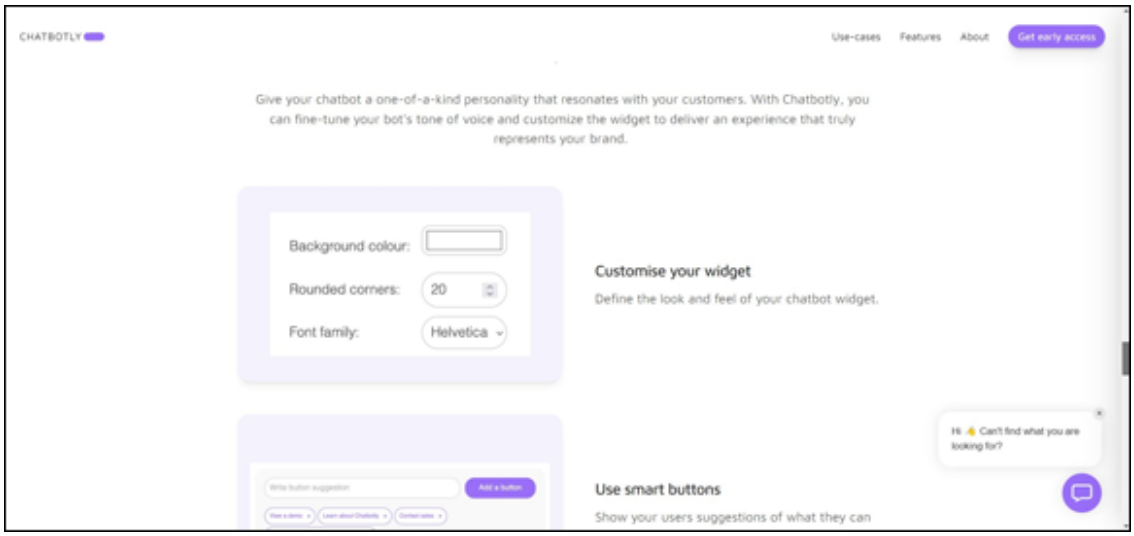


FIGURE 3-2: Chatbotly manages chatbot creation and hosting.